

Adaptation to hypoxia is a marker of a tumor's resistance to chemotherapy and radiation. The low oxygen saturation in solid tumors activates hypoxia-inducible factors (HIFs), which stimulate the expression of genes required for angiogenesis, erythropoiesis, and glucose utilization. Under normal conditions, the binding of HIF to DNA is interrupted by HIF-prolyl hydroxylases (HIF-PHDs), enzymes that hydroxylate aliphatic residues such as leucine or proline on HIFs. These hydroxylated residues serve as markers for the ubiquitin-proteasome system. Researchers have proposed that changes in oxygen-dependent pathways such as the citric acid cycle can regulate HIF-PHD activity.

Experiment 1

A kinetic study was conducted to test the effect of citric acid cycle intermediates on HIF-PHDs. Fumarate and succinate were examined with purified HIF-PHD against increasing concentrations of 2-oxoglutarate (2-OG), a cosubstrate of HIF-PHD.

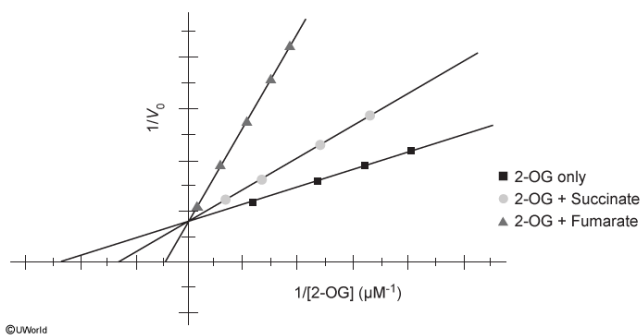


Figure 1 Activity of HIF-PHD in the presence or absence of either 0.5 mM succinate or 0.5 mM fumarate *in vitro*

Experiment 2

Researchers measured HIF-1α, a HIF subunit, using western blot analysis in wild-type (FH⁺) and fumarate hydratase-deficient (FH⁻) cells that were treated with fumarate under normoxic and hypoxic conditions (Figure 2).

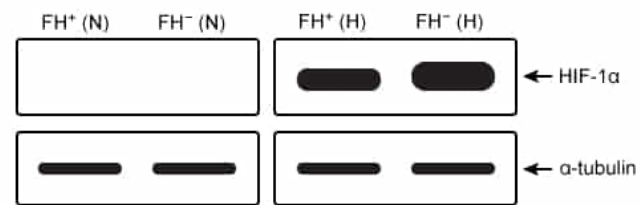


Figure 2 HIF-1α in FH⁺ and FH⁻ cells grown *in vitro* under normoxic (N) and hypoxic (H) conditions

How is the enzymatic regulation of HIF-PHDs by fumarate and succinate best described?

- ☐ A. Fumarate and succinate are equally effective noncompetitive inhibitors. [2%]
- ☐ B. Fumarate is a competitive inhibitor, and succinate is an uncompetitive inhibitor. [3%]
- ☐ C. Fumarate and succinate display varying capacity for mixed inhibition. [6%]
- ✔ ☒ D. Fumarate is a more potent competitive inhibitor than succinate. [86%]

Correct

86% answered correctly

Explanation:

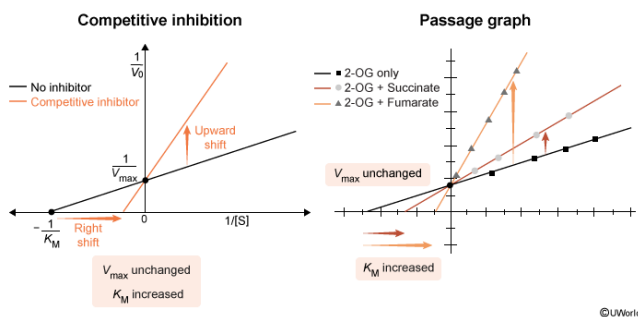


Figure 1 shows **competitive inhibition** of HIF-PHDs by fumarate and succinate on a Lineweaver-Burk plot, a graph based on a **rearrangement** of the **Michaelis-Menten equation** that is used to identify different types of enzyme inhibition. **Lineweaver-Burk plots** show the reciprocal of the substrate concentration ($1/[S]$) plotted against the reciprocal of the reaction velocity ($1/V_0$). Accordingly, the x-intercept is $-1/K_M$, the y-intercept is $1/V_{max}$, and the slope is K_M/V_{max} . K_M , a measure of *enzyme-substrate affinity*, is defined as the substrate concentration that yields half the maximum reaction velocity, or $\frac{1}{2}V_{max}$. On these plots, an increase in the apparent value of K_M is observed as a right shift of the x-intercept ($-1/K_M$), and a decreased V_{max} is shown as an upward shift of the y-intercept ($1/V_{max}$).

Figure 1 shows the kinetics of the HIF-PHD-catalyzed reaction involving 2-OG. It also shows that the inhibitors fumarate and succinate increase the K_M value

(ie, right shift $-1/K_M$) of the 2-OG reaction while leaving V_{max} unaltered (ie, $1/V_{max}$ unchanged). Fumarate and succinate can consequently be classified as **competitive inhibitors**, molecules that bind reversibly to an enzyme's active site and block substrate access, causing an **increase in K_M but no change in V_{max}** . The V_{max} remains unaltered because increased substrate concentration outcompetes the inhibitor, returning V_{max} to its original value.

Therefore, **fumarate** is a **more potent competitive inhibitor** as it results in a greater increase in the K_M value of 2-OG (ie, greater right shift) than does succinate, given equal inhibitor concentrations. Fumarate occupies the HIF-PHD active site for longer periods than succinate, effectively reducing the enzyme's affinity for 2-OG.

(Choice A) **Noncompetitive inhibitors** allosterically bind both the enzyme and enzyme-substrate (ie, ES) complex with the same affinity; they decrease V_{max} but do not change K_M .

(Choice B) **Uncompetitive inhibitors**, which allosterically bind only the ES, reduce both V_{max} and K_M by the same factor (ie, catalytic efficiency is unchanged).

(Choice C) Mixed inhibitors allosterically bind the enzyme and the ES complex with different binding affinities; this action is represented by a decrease in V_{max} and either an increase or decrease in K_M , depending on their relative affinities.

Educational objective:

The x-intercept on a Lineweaver-Burk plot represents $-1/K_M$, and the y-intercept represents $1/V_{max}$. Competitive inhibitors bind reversibly to an enzyme's active site and block substrate access, causing an increase in K_M but no change in V_{max} .

Time spent: 0 sec
Subject:
Biochemistry

QID: 400302
System:
Enzymes

Adaptation to hypoxia is a marker of a tumor's resistance to chemotherapy and radiation. The low oxygen saturation in solid tumors activates hypoxia-inducible factors (HIFs), which stimulate the expression of genes required for angiogenesis, erythropoiesis, and glucose utilization. Under normal conditions, the binding of HIF to DNA is interrupted by HIF-prolyl hydroxylases (HIF-PHDs), enzymes that hydroxylate aliphatic residues such as leucine or proline on HIFs. These hydroxylated residues serve as markers for the ubiquitin-proteasome system. Researchers have proposed that changes in oxygen-dependent pathways such as the citric acid cycle can regulate HIF-PHD activity.

Experiment 1

A kinetic study was conducted to test the effect of citric acid cycle intermediates on HIF-PHDs. Fumarate and succinate were examined with purified HIF-PHD against increasing concentrations of 2-oxoglutarate (2-OG), a cosubstrate of HIF-PHD.

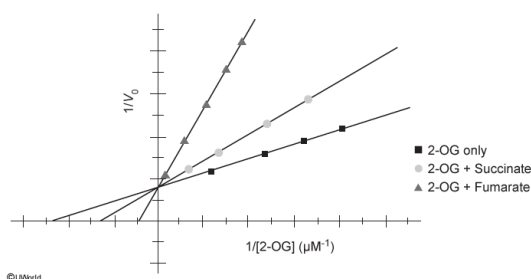


Figure 1 Activity of HIF-PHD in the presence or absence of either 0.5 mM succinate or 0.5 mM fumarate *in vitro*

Experiment 2

Researchers measured HIF-1 α , a HIF subunit, using western blot analysis in wild-type (FH⁺) and fumarate hydratase-deficient (FH⁻) cells that were treated with fumarate under normoxic and hypoxic conditions (Figure 2).

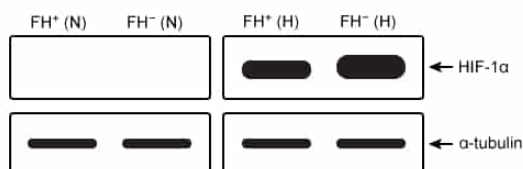


Figure 2 HIF-1 α in FH⁺ and FH⁻ cells grown *in vitro* under normoxic (N) and hypoxic (H) conditions

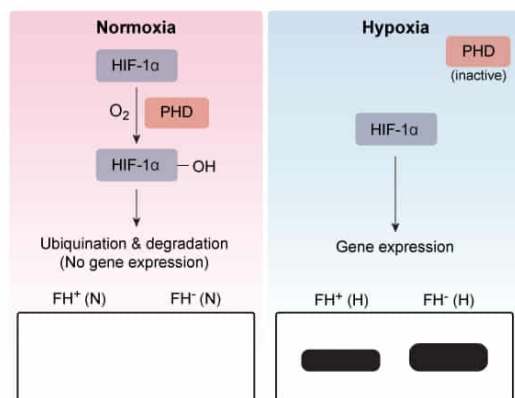
Based on the passage, culturing wild-type and FH⁻ cells in atmospheric oxygen conditions leads to:

- ☐ A. overexpression of the HIF-1 α protein subunit. [8%]
- ☐ B. reduction in activity of the HIF-PHDs. [22%]
- ☒ C. increased hydroxylation and ubiquitination of HIF-1 α . [61%]
- ☐ D. decreased proteasomal activity. [7%]

Correct

61% answered correctly

Explanation:



© UWORLD

Figure 2 illustrates the effect of hypoxia and normoxia (*normal/atmospheric oxygen levels*) on the expression of the HIF-1 α subunit in wild-type (FH⁺) and fumarate hydratase–deficient (FH[−]) cells. As expected, in hypoxic conditions both FH⁺ and FH[−] cells express high levels of HIF-1 α . In contrast, FH⁺ and FH[−] cells grown in normoxic conditions lack high levels of HIF-1 α .

According to the passage, **HIF-1 α** should be present only in **small quantities under normal oxygen** conditions because it is **hydroxylated by active HIF-PHDs**. Hydroxylation makes HIF-1 α a target for **ubiquitination**, an ATP-dependent process in which ubiquitin molecules are added to a protein substrate. The **proteasome** then **recognizes** the ubiquitinated protein (eg, HIF-1 α) and **degrades** it to small peptides via proteolysis (ie, peptide bond cleavage). As a result, HIF-1 α concentration is maximally decreased (or absent) under normoxia. In comparison, HIF-1 α concentration is high during hypoxic conditions when HIF-PHDs are inactive.

(Choice A) Normal oxygen levels induce HIF-1 α degradation (not expression) by stimulating hydroxylation by active HIF-PHDs.

(Choice B) Low (not normal) oxygen levels decrease the activity of HIF-PHDs, causing HIF-1 α levels to rise.

(Choice D) Proteasomes degrade more HIF-1 α under normal atmospheric conditions because more are ubiquitinated.

Educational objective:

Proteins are tagged for degradation in a process known as ubiquitination, which causes the proteasome to recognize and degrade the marked protein via proteolysis (peptide bond cleavage). This system is responsible for degrading damaged or unnecessary proteins/enzymes.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400303

System:
Enzymes

Adaptation to hypoxia is a marker of a tumor's resistance to chemotherapy and radiation. The low oxygen saturation in solid tumors activates hypoxia-inducible factors (HIFs), which stimulate the expression of genes required for angiogenesis, erythropoiesis, and glucose utilization. Under normal conditions, the binding of HIF to DNA is interrupted by HIF-prolyl hydroxylases (HIF-PHDs), enzymes that hydroxylate aliphatic residues such as leucine or proline on HIFs. These hydroxylated residues serve as markers for the ubiquitin-proteasome system. Researchers have proposed that changes in oxygen-dependent pathways such as the citric acid cycle can regulate HIF-PHD activity.

Experiment 1

A kinetic study was conducted to test the effect of citric acid cycle intermediates on HIF-PHDs. Fumarate and succinate were examined with purified HIF-PHD against increasing concentrations of 2-oxoglutarate (2-OG), a cosubstrate of HIF-PHD.

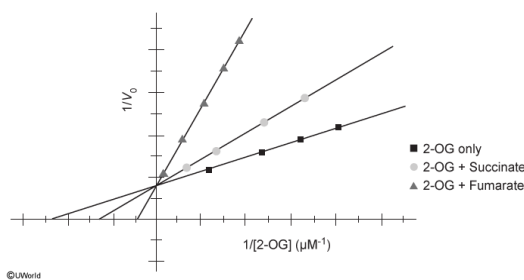


Figure 1 Activity of HIF-PHD in the presence or absence of either 0.5 mM succinate or 0.5 mM fumarate *in vitro*

Experiment 2

Researchers measured HIF-1 α , a HIF subunit, using western blot analysis in wild-type (FH⁺) and fumarate hydratase-deficient (FH⁻) cells that were treated with fumarate under normoxic and hypoxic conditions (Figure 2).

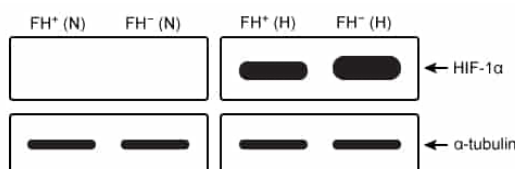


Figure 2 HIF-1 α in FH⁺ and FH⁻ cells grown *in vitro* under normoxic (N) and hypoxic (H) conditions

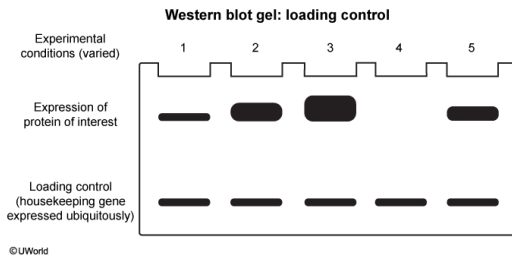
Why did researchers test for the α -tubulin protein along with HIF-1 α ?

- ☐ A. α -Tubulin expression is also generally affected by an increase in fumarate.
- ☒ B. α -Tubulin serves as a loading control because it is expressed ubiquitously. [59%]
- ☐ C. α -Tubulin is a protein that serves as an experimental negative control. [13%]
- ☐ D. α -Tubulin serves as a positive control because it is always expressed in FH⁻ cells. [26%]

Correct

58% answered correctly

Explanation:



Depicted in Figure 2, **western blotting** uses labeled antibodies to identify proteins separated by gel electrophoresis. The protein concentrations detected are represented by bands, which may vary in width/intensity corresponding to greater abundance of protein. Proteins that are **ubiquitously expressed** (ie, expressed in all cell/tissue types) and with expression levels unaffected by the experimental conditions are used as **loading controls** to show that all lanes in the gel have the same amount of loaded sample. Loading controls prevent misinterpretation of gels that have thicker bands simply because more protein was loaded into them. Properly loaded gels have the same amount of loading control in every lane.

The most common loading controls are proteins necessary for baseline cellular function transcribed from **housekeeping genes** (eg, α - and β -tubulin proteins are structural/mobile cytoskeleton components). Based on this fact, α -tubulin was used as a loading control as it can be detected at consistent levels in all cell types and has a molecular weight different from HIF-1 α . The consistent expression of α -tubulin in the gel under all conditions confirms that differences in HIF-1 α expression were not a result of experimental errors in protein loading or detection.

(Choice A) An increase in fumarate would not affect α -tubulin levels because the proteins used as loading controls are always present and remain unaffected by changes in experimental conditions.

(Choice C) A negative control is a standard experimental condition in which no response is expected (eg, using cells unable to express HIF-1 α). However, α -tubulin is always expressed, and its presence does not confirm/predict the absence of HIF-1 α .

(Choice D) A positive control is one in which the expected result is always observed (eg, using cells known to always produce HIF-1 α). α -Tubulin is not a positive control because it does not confirm/predict the presence of HIF-1 α , the expected result under hypoxic conditions.

Educational objective:

Loading controls normalize protein detection and ensure that protein loading is standardized across the gel. Proteins used as loading controls tend to be ubiquitously expressed and have consistent concentrations across all cell/tissue types. Housekeeping genes are the most common loading controls.

Time spent: 0 sec

Subject:
Biochemistry

QID: 400304

System:
Enzymes

Adaptation to hypoxia is a marker of a tumor's resistance to chemotherapy and radiation. The low oxygen saturation in solid tumors activates hypoxia-inducible factors (HIFs), which stimulate the expression of genes required for angiogenesis, erythropoiesis, and glucose utilization. Under normal conditions, the binding of HIF to DNA is interrupted by HIF-prolyl hydroxylases (HIF-PHDs), enzymes that hydroxylate aliphatic residues such as leucine or proline on HIFs. These hydroxylated residues serve as markers for the ubiquitin-proteasome system. Researchers have proposed that changes in oxygen-dependent pathways such as the citric acid cycle can regulate HIF-PHD activity.

Experiment 1

A kinetic study was conducted to test the effect of citric acid cycle intermediates on HIF-PHDs. Fumarate and succinate were examined with purified HIF-PHD against increasing concentrations of 2-oxoglutarate (2-OG), a cosubstrate of HIF-PHD.

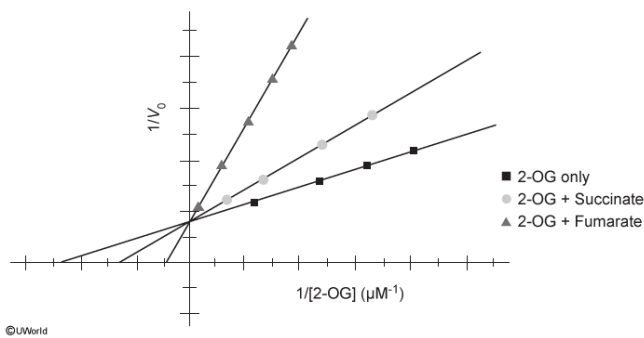


Figure 1 Activity of HIF-PHD in the presence or absence of either 0.5 mM succinate or 0.5 mM fumarate *in vitro*

Experiment 2

Researchers measured HIF-1 α , a HIF subunit, using western blot analysis in wild-type (FH⁺) and fumarate hydratase-deficient (FH⁻) cells that were treated with fumarate under normoxic and hypoxic conditions (Figure 2).

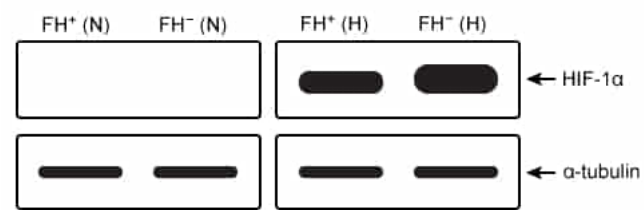


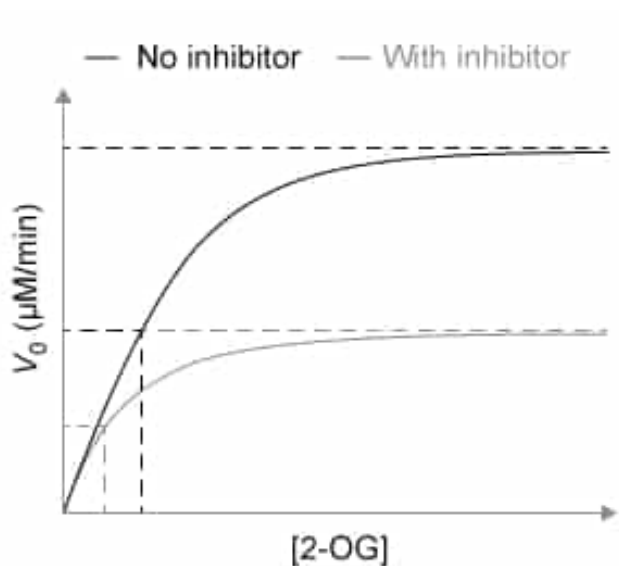
Figure 2 HIF-1 α in FH⁺ and FH⁻ cells grown *in vitro* under normoxic (N) and hypoxic (H) conditions

hypoxic (H) conditions

The peptide substrate (PS) is an inhibitor that binds HIF-PHDs outside the active site. The PS binds to both HIF-PHD and HIF-PHD–2-OG complexes with equal affinity.

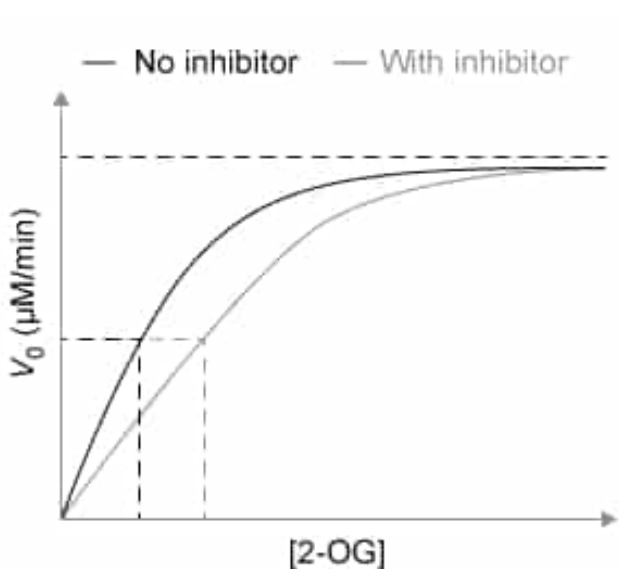
According to the passage, which graph would represent the effect of PS on HIF-PHD activity?

☐ A.



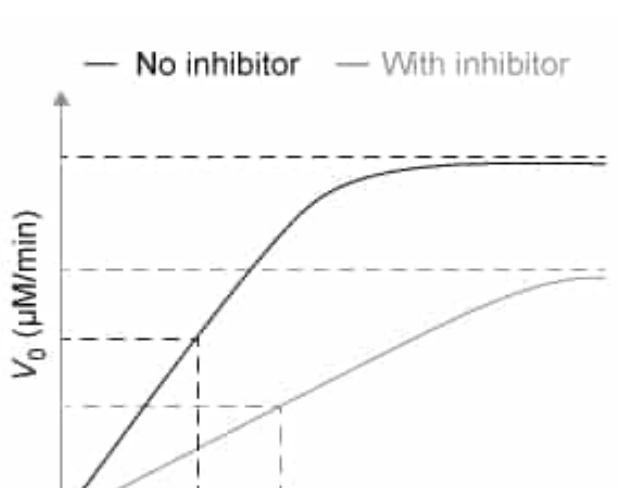
[21%]

☐ B.



[11%]

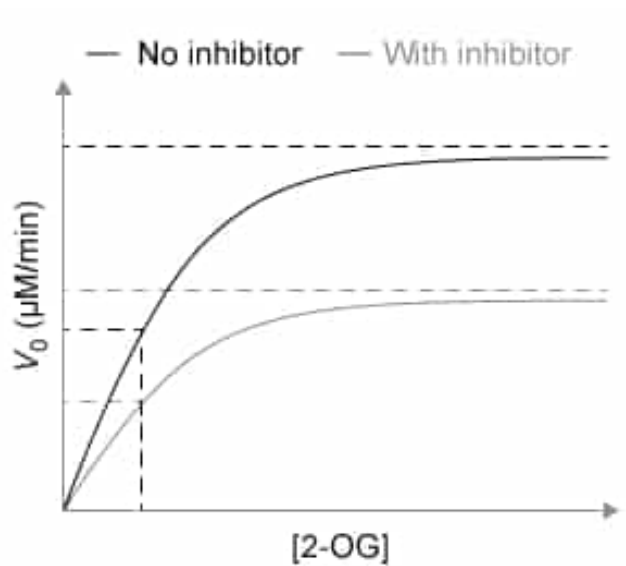
☐ C.





[7%]

✓ ☒ D.

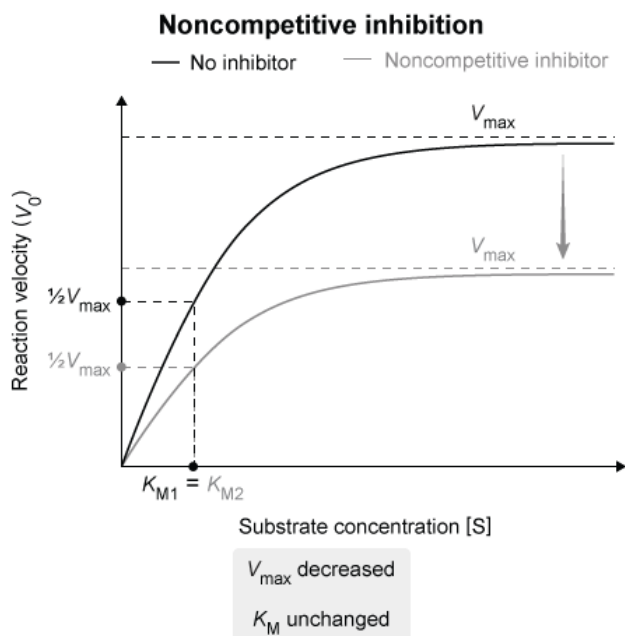


[60%]

Correct

61% answered correctly

Explanation:



Michaelis-Menten graphs plot **initial reaction velocities V_0** against **substrate concentration $[S]$** and can be used to determine $V_{\max}$ and K_M . As $[S]$ increases, V_0 also increases up to a maximum reaction velocity $V_{\max}$, resulting in a hyperbolic curve. The Michaelis constant K_M is defined as the substrate concentration where half of $V_{\max}$ is achieved (ie, $V_{\max}/2$) or where half the enzyme's active sites are full. K_M quantitatively represents the enzyme's affinity for its substrate. Changes in K_M and $V_{\max}$ can be used to identify types of enzyme inhibition.

In this scenario, the peptide substrate (PS) binds both HIF-PHD allosterically (outside the active site) and HIF-PHD–2-OG with **equal affinity**. By definition, the PS is considered a **noncompetitive inhibitor**, a molecule that **binds both the enzyme and the enzyme-substrate complex (ES)** with **equal affinity**.

Noncompetitive inhibitors slow the reaction rate and **decrease $V_{\max}$** ; however, they do not alter the affinity of the enzyme for its substrate, resulting in an **unchanged K_M** . Accordingly, in the plot of reaction velocity V_0 with increasing 2-OG cosubstrate shown in Choice D, the second curve shows that addition of the PS decreases $V_{\max}$ but does not change K_M compared to the first curve.

(Choice A) This graph shows **uncompetitive inhibition**, which decreases both $V_{\max}$ and K_M by the same factor, as these inhibitors only bind to the ES. However, the PS inhibitor binds both the enzyme and ES.

(Choice B) This graph shows an increase in K_M but no change in $V_{\max}$ when the PS is added. This represents the action of **competitive inhibitors** that bind to the active site and prevent substrate binding, an effect that can be overcome by adding more substrate, resulting in an unchanged $V_{\max}$.

(Choice C) The type of **mixed inhibition** displayed in this graph increases K_M while $V_{\max}$ decreases. Mixed inhibitors generally decrease $V_{\max}$ and have different binding affinities for the enzyme and ES at allosteric sites; as a result, they can either increase or decrease K_M .

Educational objective:

Michaelis-Menten graphs plot the initial reaction velocity against substrate concentrations and can be used to determine K_M and $V_{\max}$. Noncompetitive

inhibitors allosterically bind the enzyme and enzyme-substrate complex with equal affinity. These inhibitors decrease $V_{\max}$ but do not alter K_M .

Time spent:0 sec

Subject:
Biochemistry

QID:400305

System:
Enzymes

Adaptation to hypoxia is a marker of a tumor's resistance to chemotherapy and radiation. The low oxygen saturation in solid tumors activates hypoxia-inducible factors (HIFs), which stimulate the expression of genes required for angiogenesis, erythropoiesis, and glucose utilization. Under normal conditions, the binding of HIF to DNA is interrupted by HIF-prolyl hydroxylases (HIF-PHDs), enzymes that hydroxylate aliphatic residues such as leucine or proline on HIFs. These hydroxylated residues serve as markers for the ubiquitin-proteasome system. Researchers have proposed that changes in oxygen-dependent pathways such as the citric acid cycle can regulate HIF-PHD activity.

Experiment 1

A kinetic study was conducted to test the effect of citric acid cycle intermediates on HIF-PHDs. Fumarate and succinate were examined with purified HIF-PHD against increasing concentrations of 2-oxoglutarate (2-OG), a cosubstrate of HIF-PHD.

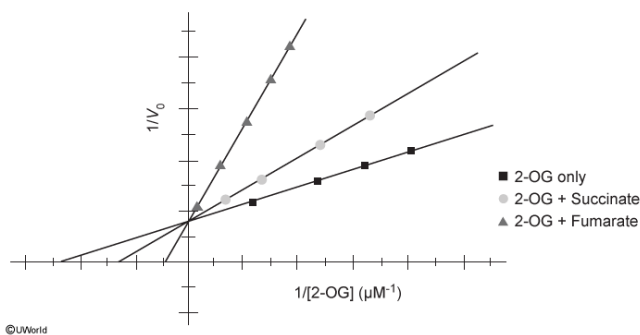


Figure 1 Activity of HIF-PHD in the presence or absence of either 0.5 mM succinate or 0.5 mM fumarate *in vitro*

Experiment 2

Researchers measured HIF-1 α , a HIF subunit, using western blot analysis in wild-type (FH⁺) and fumarate hydratase-deficient (FH⁻) cells that were treated with fumarate under normoxic and hypoxic conditions (Figure 2).

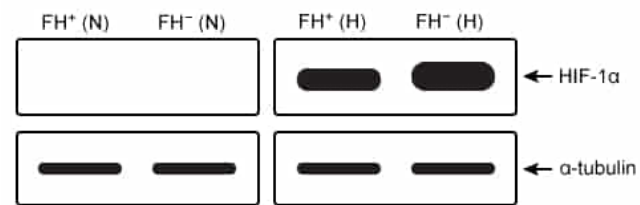


Figure 2 HIF-1 α in FH⁺ and FH⁻ cells grown *in vitro* under normoxic (N) and hypoxic (H) conditions

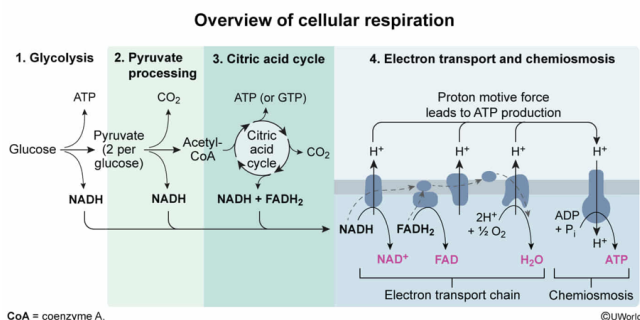
During hypoxia, skeletal muscles have been shown to accumulate succinate and ADP molecules. Which of the following explains this result?

- ✔ ☒ A. Anaerobic conditions prevent mitochondrial regeneration of NAD^+ and FAD . [54%]
- ☐ B. Succinate dehydrogenase k_{cat} is decreased after prolonged hypoxia. [32%]
- ☐ C. Excess fumarate is converted back to succinate in ATP-rich conditions. [7%]
- ☐ D. Overabundance of NADH stimulates succinyl-CoA production. [6%]

Correct

54% answered correctly

Explanation:



The accumulation of succinate and ADP during hypoxia occurs because **anaerobic conditions** prevent the **normal regeneration of the NAD^+ and FAD** needed for progression of the citric acid cycle. This cycle involves reactions that occur within the mitochondrial matrix and perform *aerobic processing* (oxidation) of pyruvate to generate CO_2 and ATP, the cell's energy currency. The **dehydrogenase enzymes** in this cycle transfer hydride ions to an electron acceptor (eg, NAD^+ , FAD), creating the high-energy electron carriers NADH and FADH_2 .

NADH and FADH_2 are then shuttled to the **electron transport chain (ETC)** in the inner mitochondrial membrane where proteins transfer the released electrons to **oxygen, the final electron acceptor**. NAD^+ and FAD are also regenerated, and the electron movement between ETC proteins generates a proton motive force that drives ATP synthase to phosphorylate ADP to ATP.

During hypoxia, less oxygen is available to accept electrons from the citric acid cycle; therefore, NAD^+ and FAD are not regenerated in the mitochondria. Consequently, ADP concentrations increase (due to the absent proton motive force) as ATP production is significantly reduced. In addition, there is an accumulation of succinate and other citric acid cycle intermediates that require NAD^+ or FAD for their oxidation reactions. Note that intermediates may not accumulate as they can leave the mitochondria (eg, citrate) or participate in side reactions like **transamination** (eg, α -ketoglutarate).

(Choice B) Succinate dehydrogenase remains fully functional during hypoxia, so its k_{cat} is unchanged. However, because FAD is a substrate of succinate dehydrogenase, hypoxia-induced depletion of FAD causes a decrease in the reaction rate.

(Choice C) Fumarate is not converted back to succinate in ATP-rich conditions because the hypoxic environment leads to a *decrease* in ATP levels. In other words, under the conditions tested, the environment is *not* rich in ATP.

(Choice D) High concentrations of NADH inhibit the α -ketoglutarate dehydrogenase that generates succinyl-CoA from α -ketoglutarate. As a result, succinyl-CoA production is reduced, not stimulated.

Educational objective:

The citric acid cycle uses oxidation reactions to convert NAD^+ and FAD to NADH and FADH_2 , respectively. The movement of electrons from these high-energy electron carriers to electron transport chain (ETC) proteins drives phosphorylation of ADP to ATP. Regeneration of NAD^+ and FAD in the ETC is only possible when oxygen is available to act as a final electron acceptor.

Time spent:0 sec

Subject:
Biochemistry

QID:400306

System:
Enzymes

Adaptation to hypoxia is a marker of a tumor's resistance to chemotherapy and radiation. The low oxygen saturation in solid tumors activates hypoxia-inducible factors (HIFs), which stimulate the expression of genes required for angiogenesis, erythropoiesis, and glucose utilization. Under normal conditions, the binding of HIF to DNA is interrupted by HIF-prolyl hydroxylases (HIF-PHDs), enzymes that hydroxylate aliphatic residues such as leucine or proline on HIFs. These hydroxylated residues serve as markers for the ubiquitin-proteasome system. Researchers have proposed that changes in oxygen-dependent pathways such as the citric acid cycle can regulate HIF-PHD activity.

Experiment 1

A kinetic study was conducted to test the effect of citric acid cycle intermediates on HIF-PHDs. Fumarate and succinate were examined with purified HIF-PHD against increasing concentrations of 2-oxoglutarate (2-OG), a cosubstrate of HIF-PHD.

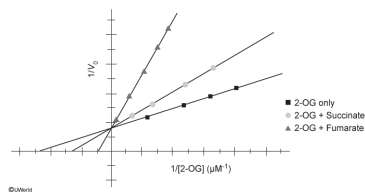


Figure 1 Activity of HIF-PHD in the presence or absence of either 0.5 mM succinate or 0.5 mM fumarate *in vitro*

Experiment 2

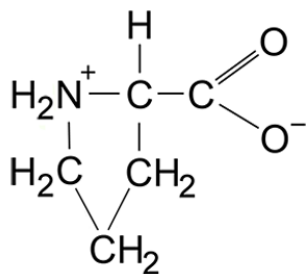
Researchers measured HIF-1 α , a HIF subunit, using western blot analysis in wild-type (FH⁺) and fumarate hydratase-deficient (FH⁻) cells that were treated with fumarate under normoxic and hypoxic conditions (Figure 2).



Figure 2 HIF-1 α in FH⁺ and FH⁻ cells grown *in vitro* under normoxic (N) and hypoxic (H) conditions

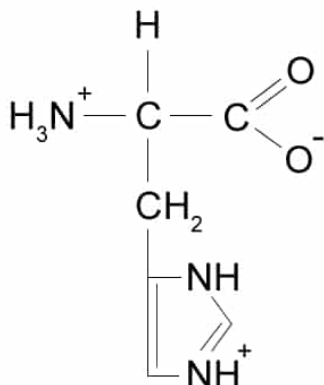
Which amino acid structure would most likely be hydroxylated by HIF-PHDs?

✓ ☒ A.



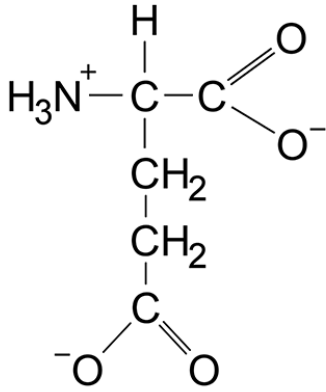
[82%]

☐ B.



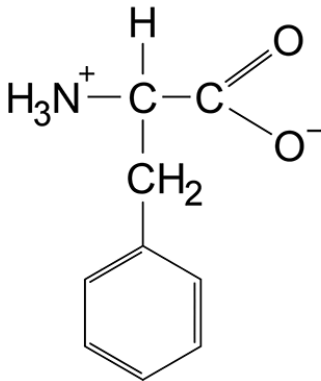
[6%]

○ C.



[7%]

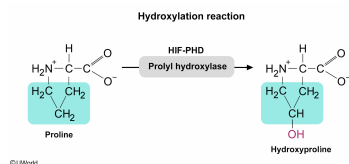
○ D.



[4%]

Correct

81% answered correctly

Explanation:

Hydroxylation reactions are involved in the oxygen-induced degradation of organic compounds and entail the **addition** of a **hydroxyl group** (–OH), usually in place of a hydrogen (H) atom. In the passage, HIF-PHD hydroxylates HIF at aliphatic residues. **Aliphatic amino acids** are **nonaromatic**, **nonpolar**, and **hydrophobic** ("water-fearing") because their R-groups are composed mostly of carbon atoms that can have single, branched, or cyclic (nonaromatic) carbon backbones.

Hydrophobicity increases as the number of carbon atoms in the side chain increases. For this reason, **glycine**, which is nonpolar but has an H atom as its R-group, is often excluded from the list of **aliphatic amino acids**. However, methionine is often included due to the inert nature of its sulfur atom. Of the choices listed, proline (shown in Choice A) is the only amino acid with an aliphatic side chain; therefore, proline is most susceptible to hydroxylation by the prolyl hydroxylase HIF-PHDs.

(Choice B) Histidine is classified as a **basic amino acid**, and its R-group can be positively charged (N-atom protonated) or uncharged at neutral pH.

(Choice C) Glutamic acid is an **acidic amino acid** that already has a carboxylic acid R-group.

(Choice D) Phenylalanine is a nonpolar amino acid but cannot be classified as aliphatic due to the aromatic ring in its R-group.

Educational objective:

Hydroxylation reactions replace hydrogen atoms with hydroxyl (–OH) groups and can occur in amino acids with aliphatic R-groups. Aliphatic amino acids are nonpolar, hydrophobic molecules with R-groups consisting of chains of C–C bonds that are single, branched, or in nonaromatic rings.

Time spent: 0 sec
 Subject:
 Biochemistry

QID: 400307
 System:
 Enzymes